MULTIPLICATION BY D ON AN ABELIAN VARIETY IS ETALE IFF D IS INVERTIBLE

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: this chapter's pool has 47 of 128 labels already delivered by the S7 line (S7-gp-* registrations, statement_anchor-prose scan; withdrawn cards excluded); none of the 11 accepted labels of this batch is among them (gp-occupied.json, 2026-10-14).

Source reference. Original lemma 0BFG: sources/stacks-chow/tags.txt, source lines 14384–14384. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0B2G: sources/stacks-chow/tags.txt, source lines 13929–13929. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GV: sources/stacks-chow/tags.txt, source lines 3242–3242. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BF5: sources/stacks-chow/tags.txt, source lines 14373–14373. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let A be a nonzero abelian variety over k. Then $[d] : A \to A$ is étale if and only if d is invertible in k.*

Proof. Observe that $[d](x + y) = [d](x) + [d](y)$. Since translation by a point is an automorphism of A, we see that the set of points where $[d] : A \to A$ is étale is either empty or equal to A (some details omitted). Thus it suffices to check whether $[d]$ is étale at the unit $e \in A(k)$. Since we know that $[d]$ is finite locally free (Lemma 0BFG) to see that it is étale at e is equivalent to proving that $\mathrm{d}[d] : T_{A/k,e} \to T_{A/k,e}$ is injective. See Varieties, Lemma 0B2G and Morphisms, Lemma 02GV. By Lemma 0BF5 we see that $\mathrm{d}[d]$ is given by multiplication by d on $T_{A/k,e}$. □

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